

Basma Ben Difallah

Laboratory of Electromechanical Systems,
National Engineering School of Sfax,
University of Sfax,
Sfax, Tunisia
e-mails: basmaben@yahoo.fr;
basma.dhifallah@gmail.com

Pierre-Henri Cornuault

Laboratory FEMTO-ST,
National Higher School of Mechanics and
Microtechniques,
University of Franche Comté,
Besançon 25000, France
e-mail: pierre-henri.cornuault@ens2m.fr

Mohamed Kharrat¹

Laboratory of Electromechanical Systems,
National Engineering School of Sfax,
University of Sfax,
Sfax 3038, Tunisia
e-mail: mohamed.kharrat@ipeis.rnu.tn

Maher Dammak

Laboratory of Electromechanical Systems,
National Engineering School of Sfax,
University of Sfax,
Sfax 3018, Tunisia
e-mail: maher.dammak@ipeis.rnu.tn

Guy Monteil

Laboratory FEMTO-ST,
National Higher School of Mechanics and
Microtechniques,
University of Franche Comté,
Besançon 25000, France
e-mail: guy.monteil@univ-fcomte.fr

On the Role of Solid Lubricant Fillers in the Tribological, Micromechanical, and Morphological Properties of PA66 Composites

To achieve thermoplastics-based composites with high tribological performances, molybdenum disulfide (MoS_2) and graphite (Gr) were incorporated separately to a polyamide 66 (PA66) matrix. Two different development methodologies of composites were used. The morphological features and the micro-mechanical properties of composites were investigated to stress the influence of the filler surface treatment and its dispersion. The friction coefficient was evaluated using a ball on a flat micro-tribometer. The wear mechanisms were also characterized. The best tribological performance was exhibited by 10 wt% of graphite. The surface treatment of the filler inclusions by 3-aminopropyltriethoxysilane (AMPTES) showed an improvement in the morphological aspects of adhesion at the filler/matrix interface. [DOI: 10.1115/1.4045804]

Keywords: thermoplastics-based composites, MoS_2 , graphite, tribological performance, micro-mechanical properties, adhesion, dry friction, wear, wear mechanisms

1 Introduction

The study of the wear behavior of polymer-based systems has attracted the attention of many researchers [1,2] owing to their wider availability of material choice, ease of manufacturing, lightweight, and good strength. The highest efficiency of such material's adoption is found in situations where sliding wear is involved [3–5]. For not long ago, the polymer-based materials were favored over their metal-based counterparts because of their low friction coefficient and capacity to sustain loads. This reason has given an impetus to the industrial manufacture of the materials as in the production of bearing components adopted in automotive industries such as cams, vacuum pumps, wheels, etc.

In this article, PA66 is selected as a target material. Polyamide 66 (PA66) composites are ideal materials for dynamic bearing applications, even at high temperatures. They are widely used in automotive applications such as automatic gears, sealing for turbocompressors pumps, pedal and gear shift components, and valve covers. The excellent mechanical properties, combined with the good wear resistance, extend their range of uses to the aerospace industry as it offers weight savings in compressor blades for turbojet engines [6].

In severe service conditions (i.e., very low or high temperature, radiation, vacuum, extreme contact pressure), solid lubricants are the best choice to control the friction and wear of thermoplastics.

When present at a sliding contact, solid lubricants act in the same way as their liquid counterparts. Particularly, they shear easily to prevent wear damage in the sliding interface and to reduce the friction coefficient [7–9]. Several inorganic materials (e.g., graphite, molybdenum disulfide “ MoS_2 ,” boric acid, hexagonal boron nitride) can provide excellent anti-friction and wear ability. Their lamellar or layered structure confers lubricity to most of these solids. Recent advances indicated that the black phosphorus (BP) was selected as a new two-dimensional material with intriguing mechanical and lubrication properties. The incorporation of BP nanosheets into organic (polyetheretherketone (PEEK)/polytetrafluoroethylene (PTFE)) composites and inorganic (carbon fiber/PTFE) composites could reduce friction and wear of both systems [10]. For BP/polyvinylidene difluoride (PVDF) composites, it was reported that BP content larger than 10 wt% enhanced the friction properties and the wear resistance of the composites [11]. Puértolas et al. [12] studied the lubricant ability of graphene nanoplatelets (GNPs) in improving the PEEK wear properties. The GNPs were used without any chemical functionalization in thermoplastic-based materials, the results showed an enhancement in the surface hardness and in the tribological properties. Zhang et al. [13] reported that aramid particles played a synergetic effect with sulfur dioxide (SiO_2) and PTFE nanoparticles in improving the tribological performance of polyoxymethylene (POM) composites.

Graphite and MoS_2 are natural minerals with lamellar structure extracted from deposits around the world. The lubricity of graphite lasts up to 400 °C, while MoS_2 decomposes at about 200 °C. Recently, partially graphitized carbon composites are developed

¹Corresponding author.

Contributed by the Tribology Division of ASME for publication in the JOURNAL OF TRIBOLOGY. Manuscript received June 27, 2019; final manuscript received December 10, 2019; published online December 20, 2019. Assoc. Editor: Yonggang Meng.

to achieve better lubricity at elevated temperatures. On the other hand, MoS₂ has the best overall load-carrying capacity as films deposited on rigid substrates and it is considered among the most suited for aerospace and cryogenic applications [14].

Nevertheless, the deployment of particle-reinforced polymer matrix composites as components for use in real service needs a good understanding of the processing associated with the structure and its consequence on physical and mechanical properties. In this field, most recent research studies have provided a theoretical guide and experimental tools to specifically tailor the matrix/filler morphology and surface chemistry in order to improve their adhesion and decline the propensity for agglomeration [15,16]. Wu et al. [17] have chemically treated the graphite powders to make them well compatibility with the polyamide 6 matrix and they studied the mechanical and electrical properties of the resultant composites. The bending and tensile strengths were improved by the incorporation of a modified graphite powder. Although the tribological properties of thermoplastic composites filled with different reinforcing fillers have been extensively studied, the effect of the development methodology and the filler's surface treatment on micro-mechanical and morphological properties was not systematically compared yet.

The purpose of this study is to investigate the effect of adding untreated and treated MoS₂ and graphite with 3-aminopropyltriethoxysilane (AMPTES) to a PA66 matrix. In this context, two development methodologies are proposed. The tribological behavior of PA-based composites is reported. Micro-mechanical characterizations and morphological aspects of the elaborated composites are also studied in order to stress the effect of the processing methodology on composites properties.

2 Experimental Details

2.1 Materials. The composites investigated in the present study consisted of two flaky shaped solid lubricant powders incorporated separately PA66 granules as the matrix material. The polymer was provided by Mapex (1.14 g/cm³ density, 250 °C melting point) and dried for 3 h at 85 °C.

The used filler materials were MoS₂ powder (6 μm average size, 5.06 g/cm³ density) with 95% purity and graphite powder of average particle size in the range of 22 μm (1.9 g/cm³ density). The two materials were obtained from Aldrich Chemistry (Sigma-Aldrich; Figs. 1(a) and 1(b)).

Different compositions were considered in this work in the range of 2, 4, 7, and 10 wt% graphite or MoS₂.

2.2 Composites Preparation. The composites preparation was achieved with two methodologies. Concerning the first conventional methodology (M1), specimens were produced by direct injection molding using Demag Ergotech 125/475 industrial injection-molding machine. The Shore A hardness of neat PA specimens was 94.66 ± 0.57 and Ra (the average roughness) was about 0.42 μm.

As for the second methodology (M2), solid lubricant powders were chemically treated. The surface treatment of MoS₂ particles was carried out in a suspension containing 5 g MoS₂, 10 ml ethanol, and 0.05 ml 3-aminopropyltriethoxy silane (AMPTES) (Sigma-Aldrich, 0.949 g/ml density). The suspension was stirred mechanically (RZR1, Heidolph) for 45 min, then dried by vacuum evaporation for 3 h. With respect to the surface treatment of graphite, it was achieved by stirring 5 g graphite with 25 ml ethanol and 0.015 ml AMPTES. The stirred solution was vacuum-dried for 6 h. The blending of the polymer with varying weight contents of treated solid lubricants was achieved by mechanical mixing using an internal mixer with two blade impellers. Thereafter, specimens were produced using an Arburg Allrounder 220 S injection-molding machine, the surface roughness of neat PA_M2 was about $Ra = 0.36$ and the Shore A hardness was 92.00 ± 1.63 . The reason for using the two methodologies is to compare the industrial manufacturing process of composites by using direct injection molding via Demag Ergotech industrial injection machine with

the second development methodology at the laboratory scale including surface treatment of the filler and mechanical mixing with the polymer. The Arburg Allrounder injection-molding machine provides processing with small shot weights.

2.3 Friction and Wear Testing. A reciprocating ball on a flat micro-tribometer was used for friction and wear tests. The description of the homemade device was published in a previous research work [7]. A mechanical arrangement generated the required contact load. The magnitude of the dead load can be adjusted to very low values. The tangential load was monitored by a measuring head which was equilibrated by a counterbalancing arm. The system was based upon a translation table including precise linear ball bearing and enabling the micro-displacement of the flat specimens. In order to avoid any mechanical vibrations of the system, all the mechanical elements were fixed on an inertial marble structure with damping devices. A computer based software ensured the treatment of the measurements and the display of the results.

A high-chromium steel ball with a diameter of 19.0 mm and an initial surface roughness of $Ra = 0.12 \mu\text{m}$ served as a counterpart. The ball slides reciprocally against the flat composites at a frequency of 1 Hz over a displacement magnitude of 2 mm. The tribological investigations were performed with a constant normal load of $F_N = 5 \text{ N}$ up to 10,000 sliding cycles (peak Hertzian contact pressure in the ball on flat contact of about 35 MPa). For a better comparison between the different composites, at least three friction tests were conducted for each set of conditions.

The wear scar shape was measured with an Infinite-Focus Alicona system at the end of each test. Before wear volume measurement, the samples were ultrasonically cleaned to eliminate the wear debris.

In order to analyze wear mechanisms, scanning electron microscope (SEM) observations (JEOL JSM 6400F) were also conducted on the wear track coated with gold.

2.4 Micro-Mechanical Characterization: Scratch Test.

Polymer scratch performance has drawn significant interest in recent years. Particularly, the scratch behavior of thermoplastic composites was investigated in previous works [18–23]. In this context, it was reported that the structure-property relationship between morphology and scratch behavior in the case of polypropylene showed ironing and surface plowing mechanisms in the deformation mode [23–25]. It is hypothesized that the progression of scratch damage mechanisms is related to the fish scale and plowing transitions. The first transition is marked by the appearance of fluctuations and a gradual increase in the tangential force measured by the scratch machine. In the second transition, abrupt changes and large waves were seen in the load profile [21,26].

The micro-mechanical analyses of composites were carried out by means of the Revetest Automatic Scratch Tester (Institut Jean Lamour-Département N2EV) which performs scratches under either constant or linearly increasing load.

Tests were drawn over samples using a sphero-conical diamond indenter of 120 deg cone angle and 138 μm radius spherical tip, under continually increasing load. The loading initial value was 1 N, the rate was 14.5 N/min, and the scratching speed was 2.5 mm/min. The indenter is moved over the sample surface with a linearly increasing load until a normal load of 30 N is reached (Fig. 2). Tangential friction force, friction coefficient, penetration, and residual depth were continually monitored. The abrupt increase in the load is termed the “critical load” for the onset of scratch visibility; this critical value was used to study the material plowing resistance [21,24,27,28].

2.5 Microstructural Characterizations

2.5.1 Fractography. Fractography analyses were established to interpret the features observed on fractured surfaces and thus to identify the fundamental failure modes. As new composites have been developed, this technique is considered as a reliable tool for

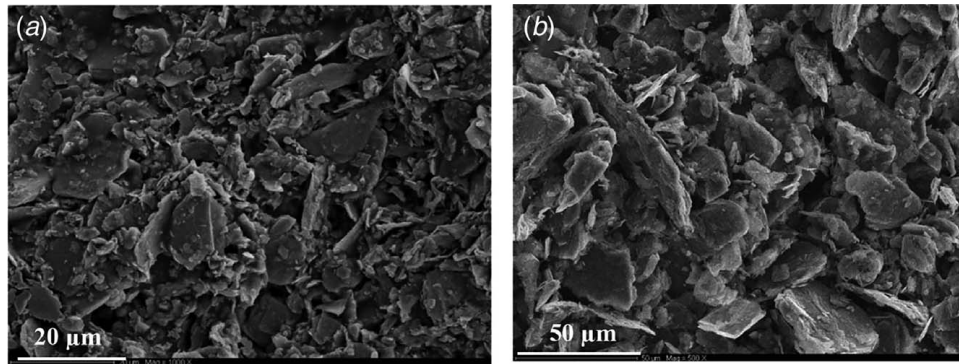


Fig. 1 SEM micrograph of (a) MoS₂ powder and (b) graphite powder, as supplied

researchers to relate fractographic morphologies to the failure processes in a component [29]. In fact, composite fracture surface analysis is directly affected by the mechanical strength in the matrix/filler interface. In our study, valuable information was provided by this microstructural characterization. The fractographic observations were provided with a SEM (JEOL JSM 6400 F) and fracture surfaces were gold-coated before use.

2.5.2 X-Ray Tomography. X-ray tomography makes use of the refraction of X-rays by the specimen in image formation. Image processing relies entirely on X-ray absorption by the analyzed specimen as a mechanism of contrast formation. These types of phase-contrast imaging methods warrant the visualization of details in weakly absorbing materials such as polymers or objects that do not absorb X-rays at all. In the context of materials science, X-ray phase-contrast imaging has become an important technique for the 2D/3D construction of low-density materials. It allows material characterization in terms of voids and cracks detection. It also determines particle size and spatial distribution of the filler. Hence, the X-ray computed tomography is considered as a reliable tool to characterize the performance of our composites.

The principal experimental parameters of the Phoenix-X-ray device used (vto me x-S) are an X-ray tube voltage of 80 kV, a scan time of approximately 1 h, a silicon-based detector is amorphous with 1920 × 1536 pixels of size, and the resolution adopted is 3 μm.

3 Results and Discussion

3.1 Friction and Wear Behaviors of the Composites

3.1.1 Friction Testing Results. Figure 3 shows the friction coefficient of typical PA66 composites filled with solid lubricant

powders by the two development methodologies as a function of sliding cycles (total sliding cycles of 10,000). These composites contain different concentrations of solid lubricant powders. The frictional behavior is to some extent reflected in this friction data. The friction coefficients of composites with 10 wt% Gr prepared with M1 and with M2 achieve a relatively stable steady-state after a short running-in phase [7]. However, the friction coefficient values did not stabilize as a function of the sliding cycles and showed a continuously increasing trend for most of the tested specimens. For pure PA, an increase in friction coefficient is seen with the second methodology. It can also be seen that the friction coefficient decreases as a function of the graphite content with both methodologies.

In Fig. 4, friction coefficients measured after 10,000 sliding cycles for all PA composites are given. Meanwhile, the composites manufactured by direct injection molding give rise to the lowest friction coefficient. Interestingly, the composite with 10 wt% of graphite exhibits the lowest coefficient of friction which falls from 0.19 with pure PA to 0.06. At the same graphite content, the friction coefficient is 0.16 for the composites manufactured by the second methodology. In the case of PA/MoS₂ composites injected and molded directly, the frictional coefficient decreased slightly as a function of the solid lubricant content up to 7 wt% after which it increased. This is in agreement with the results published by Wang et al., who filled metakaolinite-based geopolymer with PTFE particles [30]. An optimum filler fraction contributes to the shearing force's reduction between the surfaces in contact, leading to a better friction response. As compared with direct injection molding, the frictional coefficient of PA/MoS₂ composites can further be reduced by the second methodology at 10 wt% of the filler. In the present case, PA composite filled with graphite flakes and injected molded directly demonstrated the best improvement in the friction performance.

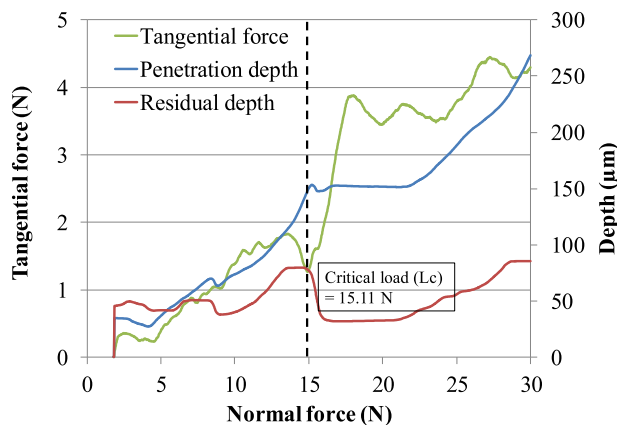


Fig. 2 Typical experimental plots for scratch adhesion tests on tests on PA/Gr composite

3.1.2 Wear Test Results. Figure 5 shows the changes in the wear volume loss with respect to the composite's preparation method, nature, and the proportion of the incorporated filler into the PA matrix. An overall decreasing trend of the wear volume loss can be observed with the increase in wt% of the solid lubricant filler. The decreasing tendencies obviously differ for each different condition. The best result is exhibited by the composition with 10 wt% of graphite, with a wear volume loss of $11.79 \times 10^{-3} \text{ mm}^3$, which is about six times lower than the value measured for the pure PA. It is well known that graphite can be easily dragged out from the matrix to form a third body transfer film dispersed on the sliding counterparts [31,32]. The quantity of graphite on the sliding surface increases directly with its amount added to the composites. The plentiful graphite flakes spread symmetrically on the sliding surface, which decreases the direct contact between the composites and the steel counterpart [33]. It is also worth pointing out that wear volume loss of composites was higher with the second manufacturing methodology that may be attributed to the processing conditions (including mechanical mixing under electric

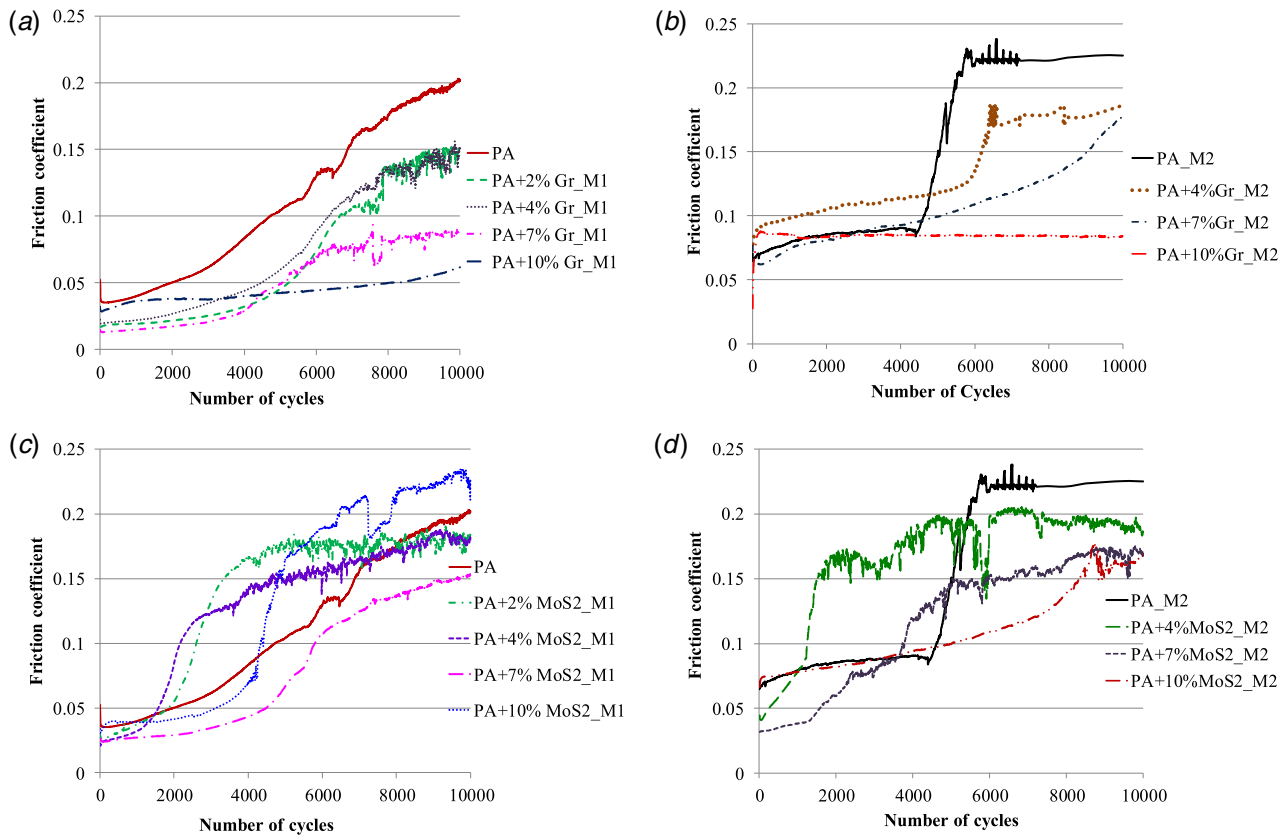


Fig. 3 Typical evolutions of friction coefficient plotted against the number of sliding cycles for (a) and (b) PA/Gr composites prepared by M1 and M2, respectively, and (c) and (d) PA/MoS₂ composites prepared by M1 and M2, respectively

resistance heaters for the second methodology) and the use of two different processes setting specific to the injection units.

3.1.3 Analysis of Wear Mechanisms. Table 1 shows the optical 3D surface measurements of the wear track's central part for pure PA and selected PA composites elaborated by direct injection molding using a confocal scanning optical microscope (Sti@1 sensor type). The 2D profile in the middle of the wear track was also determined. A considerable difference in the topography is noticed between the pure PA and PA with 7 wt% MoS₂ and 10 wt% Gr composites. The PA surface profile appears to be rough and jagged, while the wear picks in the case of PA/MoS₂ composite profile are relatively stumped, a micro-plowing wear

mechanism is activated. For the PA with 10 wt% Gr composites, it is worth mentioning that the wear profile was difficult to measure because it became fairly flat and regular. Thus, wear was extremely reduced when adding 10 wt% of graphite.

The SEM analyses shown in Figs. 6 and 7 were performed on the wear scars and the pin worn surfaces of (a) neat PA66, (b) PA with 7 wt% MoS₂, and (c) PA with 10 wt% Gr, respectively. It is to be noted that depending on the filler nature and proportion, the worn surfaces experienced different wear mechanisms. The wear track of pure PA (Fig. 6(a)) was rough and characterized by severe scuffing. At higher magnifications (insets), the evidence of plastic deformations and particle detachment were observed in the micrographs. In the course of the periodic sliding process, the repeated plowing

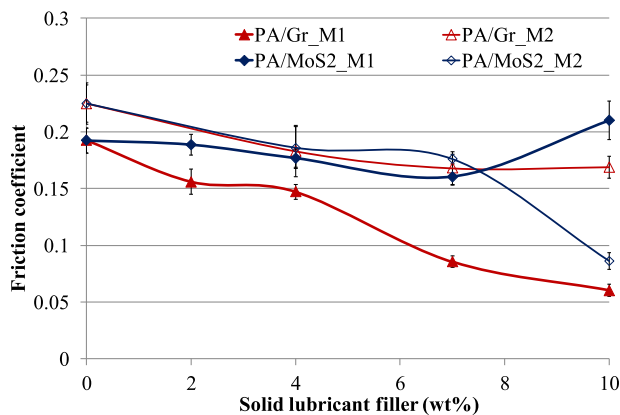


Fig. 4 Variation of friction coefficient after 10,000 sliding cycles for PA composites as a function of the specimen preparation and composition

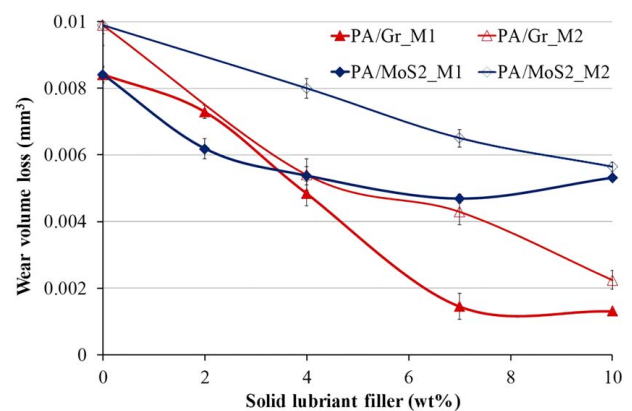
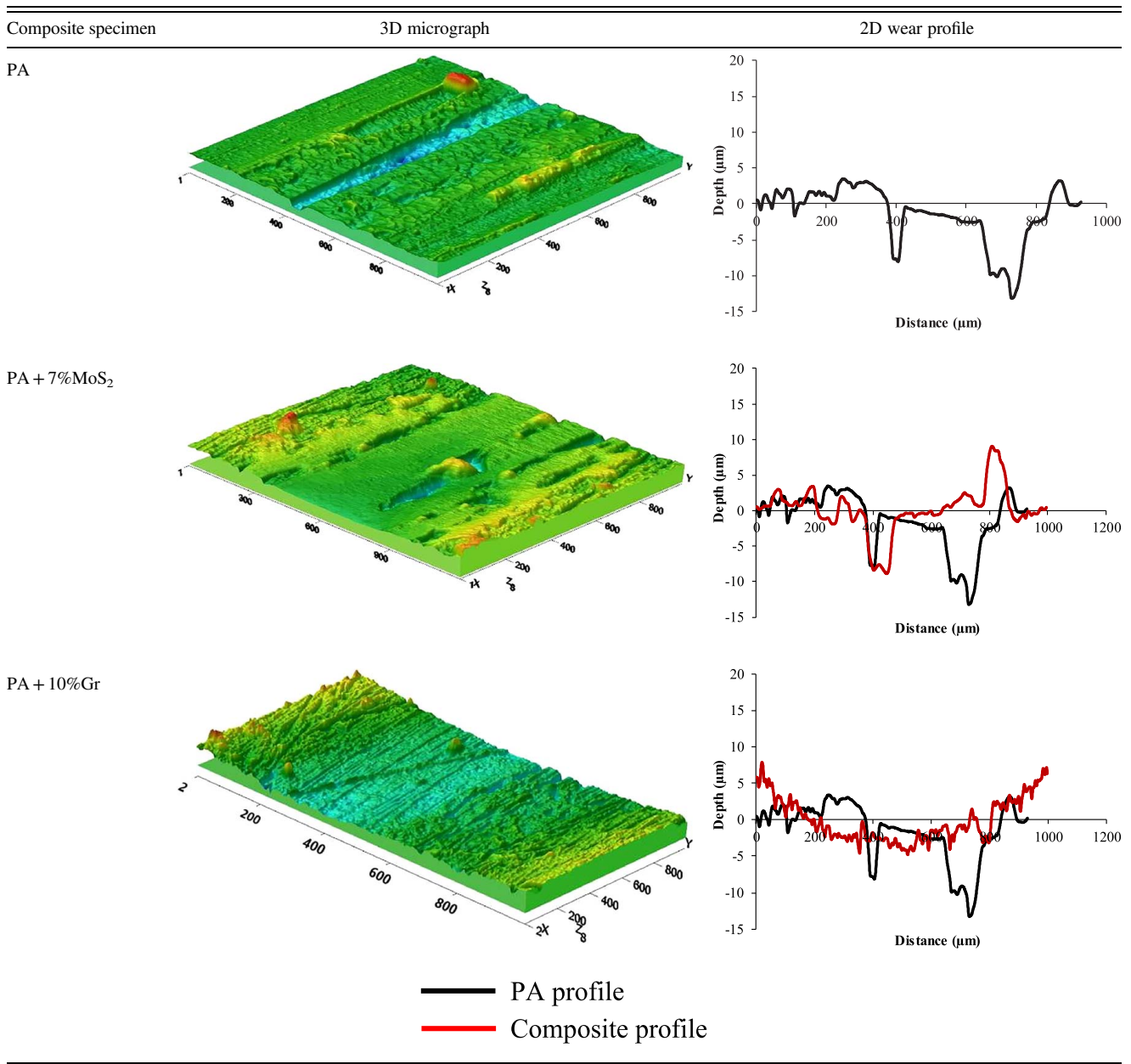


Fig. 5 Comparison of wear volume loss of PA composites with the variation of the nature and the wt% of the solid lubricant filler for the two development methodologies, 10,000 sliding cycles

Table 1 Typical three-dimensional images and transversal profiles recorded on the worn surfaces of neat PA and PA composites injected molded directly



mechanism caused the surface fatigue, thus debris was pulled out from the polymer. The counterpart steel pin (Fig. 7(a)) reveals the existence of abundant wear particles detached from the polymer and electrostatically deposited on the pin surface under mechanical interactions between the surfaces in contact. Several previous studies proved the PA aptitude in forming a uniform and adherent transfer film on the steel counterpart [34–36]. However, Samyn et al. have demonstrated that the contact between the PA specimen and its steel counterpart depends on the time-load temperature related to overload conditions, and hence the interference with softening and/or melting for PA [37]. For contact temperatures below the glass transition temperature, wear debris accumulated in the sliding stroke and did not compact in a transfer film. In our case, the worn surface of PA indicates a severe adhesive mode in a softening regime.

In the case of PA with 7 wt% MoS₂, the composite wear track is quite smooth when compared with pure PA. At higher magnifications (Fig. 6(b)), it can be seen that the composites become

plastically deformed under the repeated plowing mechanism. Indeed, the microcracks that were formed along the deformed material were pulled out by brittle fracture. The SEM image of the transfer film formed on the counterpart pin (Fig. 7(b)) shows discontinuous and thick transfer. A large number of transferred PA debris is also observed on the high-chromium steel counterpart. The reasons for the above result may be as follows. Concerning the first suggestion, it relates that the sliding test was interrupted before the transformation of the compacted debris in an adherent and coherent transfer film [38]. As for the second suggestion, it pertains to the fact that during the sliding process, the transfer film was damaged by the newly formed wear particles [39]. In all cases, the addition of MoS₂ flakes to PA66 promoted the development of a third body with a lamellar structure, which provided a better anti-wear property for the corresponding composite.

Figure 6(c) shows the SEM micrograph of the worn surface of 10 wt% graphite filled PA. It can be seen that the composite worn surface was smooth and only exhibited gentle micro-plowing

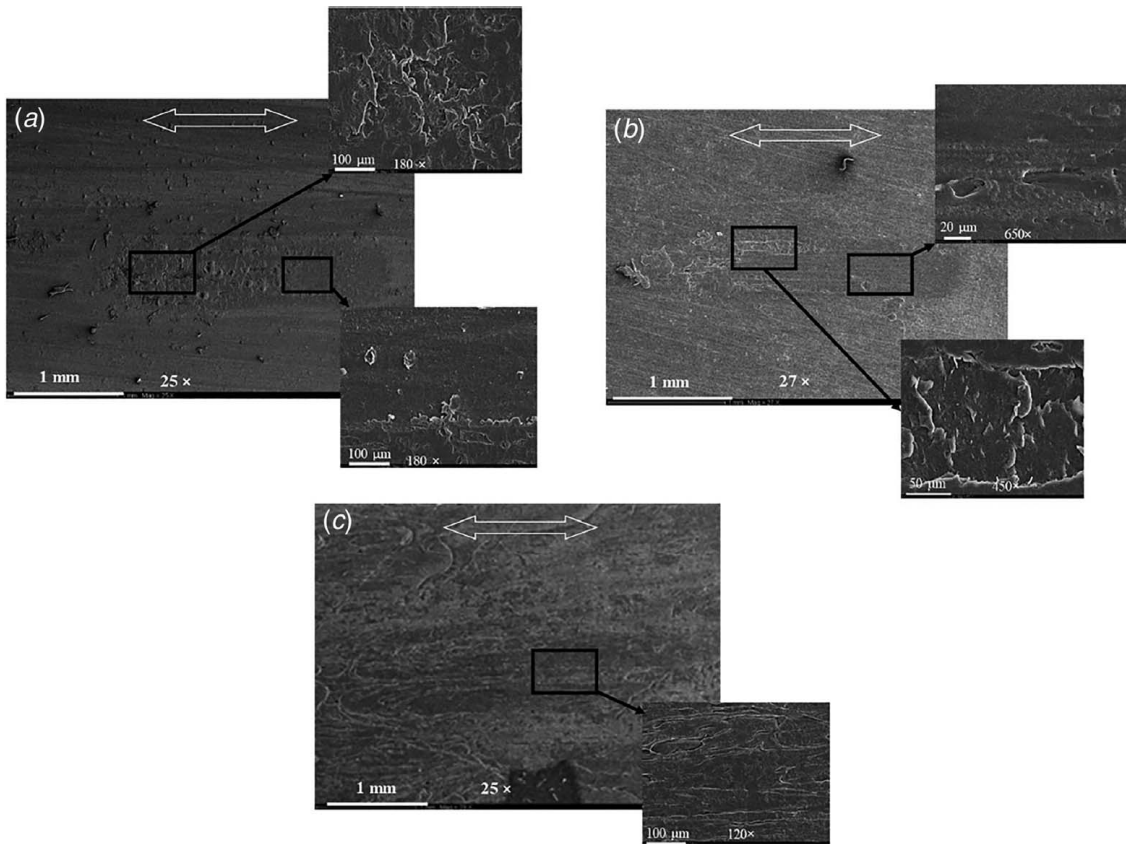


Fig. 6 SEM of the worn surfaces after sliding tests for (a) unfilled PA, (b) 7 wt% MoS₂ filled PA, and (c) 10 wt% graphite filled PA, arrows indicate sliding direction

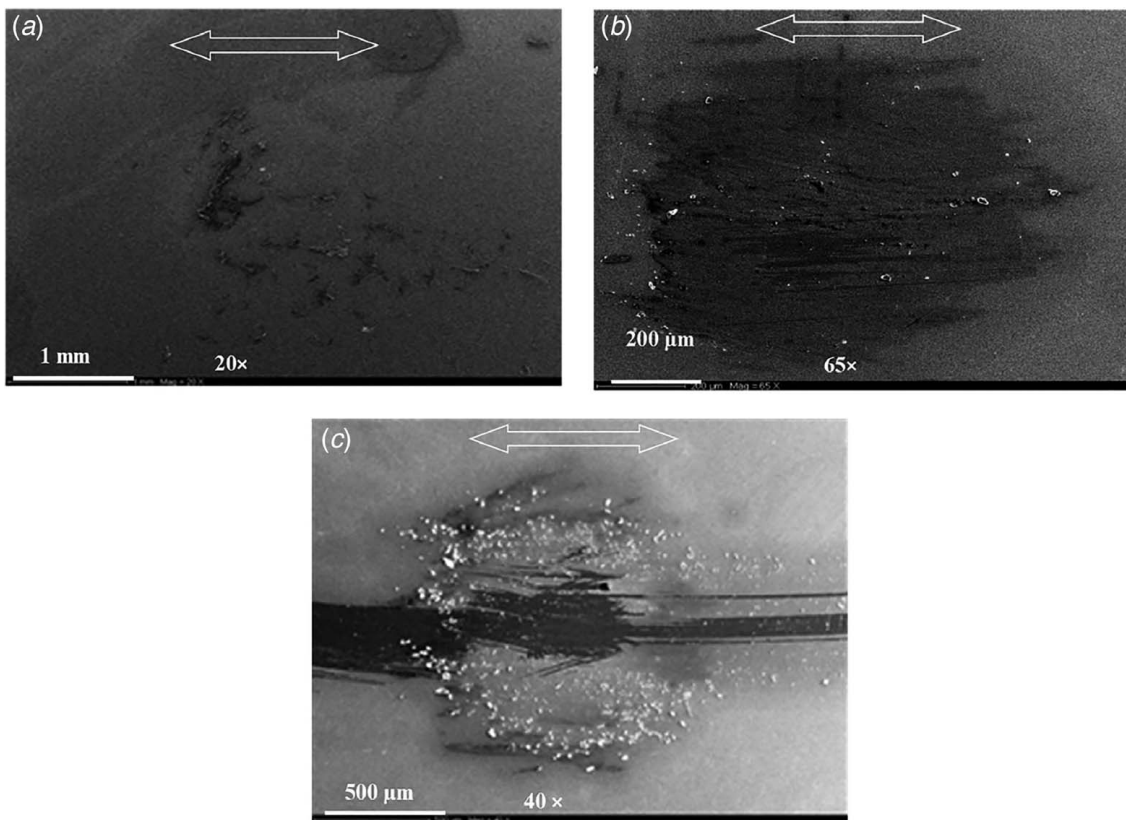


Fig. 7 SEM of the pin worn surfaces after sliding tests for (a) unfilled PA, (b) 7 wt% MoS₂ filled PA, and (c) 10 wt% graphite filled PA, arrows indicate sliding direction

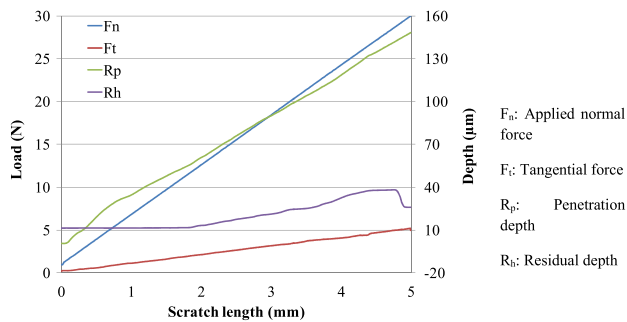


Fig. 8 Typical scratch test plot for pure PA

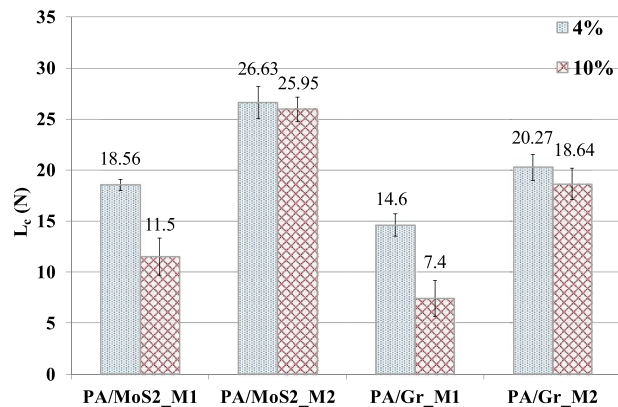


Fig. 9 Comparative chart of critical load, L_c , after scratch testing for PA composites (M1 indicates the first conventional methodology of preparation and M2 indicates the second preparation methodology)

phenomena. The magnified detail shows no obvious particle detachment phenomena. As can be seen in Fig. 7(c) showing the steel counterpart, it appears that a transfer film has been created (Fig. 7(c)) by the presence of a large amount of powder covering the counterface completely. These transferred particles are expected to be the compacted debris sticking together to form a uniform film on the substrate surface. The surface reactivity seems to be higher with the addition of graphite particles that led to a significant decrease in the wear volume loss and in the friction coefficient as well.

3.2 Scratch Test Results. Figure 8 illustrates the typical results of the scratch test performed on pure PA and plots applied force (left side) and vertical displacement (right side) versus scratch distance. This figure reveals that the tangential force curve does not present a neck during the scratch test. Hence, there is no point in the curves that may identify a clear onset of failure, which can be explained by the absence of any ductile plow scratch damage with increasing load [40]. It can also be suggested

that the polymer was progressively damaged during the scratch test [21].

Figure 9 shows the typical critical load of four composites (PA/Gr_M1, PA/Gr_M2, PA/MoS₂_M1, and PA/MoS₂_M2) at 4 wt% and 10 wt%, respectively, whereby each histogram has an average of 10 different measurements. It was reported that the improved scratch resistance of thermoplastic polymers leads to higher critical loads [24]. As can be seen from the figure, the PA/Gr composites elaborated with the first methodology exhibit the minimum values of the critical load, the average values are significantly improved with the second methodology. At 10 wt% of graphite, the critical load rises from 7.4 N to 18.6 N. The load values decrease as the solid lubricant percentage increases. It was proved that undesirable failure mechanisms present during scratching are evident points strongly related to the role of the filler particles in the scratch behavior [21,25]. It was also suggested that the critical load is related to fracture criteria and some yielding as a specific failure mode [23]. Debonding of the solid lubricant particles and the formation of voids would probably lead to the decreasing tendencies in the scratch load as a function of the filler content [21,23]. Wong et al. showed that talc-filled PP was found to be responsible for the great increased scratch visibility of PP due to the massive formation of voids as well as crack and debonding phenomena on the scratched surface [21]. The best results of the critical load are given with PA/MoS₂ elaborated with the second manufacturing methodology.

Table 2 shows the residual depth Rh , the penetration depth Rp , and the viscoelastic recovery parameters under the highest applied load. It is interesting to notice here that the penetration depth reveals the real-time deformation behavior of the composite sample during the test, while the residual deformation data records the plastic deformation induced by the indenter. The difference between the two depths may indicate the ratio of viscoelastic deformation under loading conditions; the larger the difference is, the larger the viscoelastic component is [26]. The surface damage is controlled by the plastic deformation experienced by the sample [26]. The percentage of viscoelastic recovery f can be calculated as [20]

$$f = [1 - (Rh/Rp)] \times 100 \quad (1)$$

The best scratch resistance parameters are recorded for neat PA, while scratches of non-treated graphite filled composites are extremely deeper. As expected, as the filler fraction increases, the depth values increase. The viscoelastic recovery is improved for both cases of filler with the surface modification process, the residual depth values are also reduced. In accordance with the critical load, PA/MoS₂_M2 exhibits good scratch resistance parameters with high viscoelastic recovery and less residual depth of the material. The residual deformation of PA/Gr_M1 occurs at high depth values even though an improvement is seen with the second methodology.

It is clear that the presence of solid lubricant particles hampers the viscoelastic recovery of the polymer chains. However, more resistance to the conical indenter (shallower penetration depth) and higher recovery are registered after surface properties modification

Table 2 Scratch resistance parameters of PA composites: effect of the filler fraction and the processing methodology

	1st methodology					2nd methodology				
	PA66	wt% Gr		wt% MoS ₂		PA66	wt% Gr		wt% MoS ₂	
		4	10	4	10		4	10	4	10
Rp (μm)	149.3	238.7	252.6	155.1	170.7	167	175.4	146.1	136.8	
Rh (μm)	25	74.5	81.2	39.7	48.9	49.8	66.2	31.4	35.9	
f (%)	83.2	68.7	67.8	74.4	71.3	70.1	62.2	78.5	73.7	

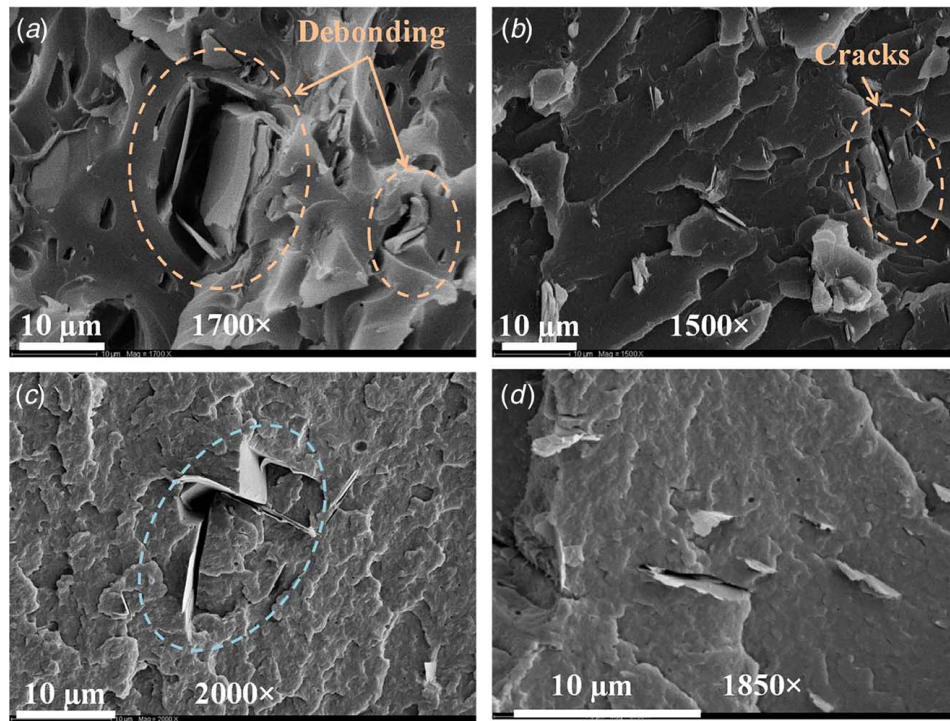


Fig. 10 SEM micrographs of fractured PA composites containing (a) 10 wt% of untreated MoS₂ (M1), (b) 10 wt% of untreated graphite (M1), (c) 10 wt% of treated MoS₂ (M2), and (d) 10 wt% of treated graphite (M2)

of the filler particles. It is reasoned that the improved interfacial adhesion of the solid lubricant particles will increase the scratch performance of the composites [21,23]. It is also proposed that the reduction of particle size and the increased stiffness and yield strength of thermoplastic polymers reduces the scratch visibility and then improves the scratch behavior of polymers [21].

3.3 Fractography Analysis. Matrix fracture is used in order to qualitatively assess the microscopical features of the composites after a brittle fracture. In Fig. 10, the SEM micrographs recorded on the fracture surface of the PA sample with 10 wt% of the filler are presented. The particles in Figs. 10(a) and 10(b) are untreated with the coupling agent, the adhesion is clearly weak and many signs of debonding, cracks, and particle detachment are visible. Figures 10(c) and 10(d) show fractured surfaces of PA composites in the case of particles coated with AMPTES, indicating that the adhesion between the components is improved. The filler is covered almost completely by the PA matrix. The silane employed as a coupling agent is able to form strong chemical bonding between the filler and the PA matrix [41,42], it establishes molecular bridge leading to a better interfacial adhesion and improved micro-mechanical properties of the composites.

3.4 X-Ray Tomography Results. Figure 11 show the typical X-ray images obtained for PA composites at 10 wt% MoS₂ developed with the first and the second methodologies. The results given in Figs. 11(a) and 11(c) bring us enough information about porosity level and size and filler distribution in the PA 6-6 based composites with untreated (M1) and treated MoS₂ (M2), respectively. The comparison of the two development methodologies reveals how the first methodology induces a large average pore number in a great abundance. Voids cover the particles of solid lubricant powder, which explains the poor adhesion between the filler and the matrix. The three-dimensional views in Figs. 11(b) and 11(d) expose a highly dispersed morphology of MoS₂ particles

in the composite prepared using the second development methodology relative to the dispersion using the first one.

The X-ray tomography exhibits the full potential to recreate three-dimensional images of the bulk volume of our composites, revealing the distinct spatial distribution of the filler. The effective dispersion of the fillers in the thermoplastic matrix and the improvement of filler-polymer interactions are two key challenges in the present work. The mechanical mixing in the second process thus shows improvement in the dispersion of filler in the polymer

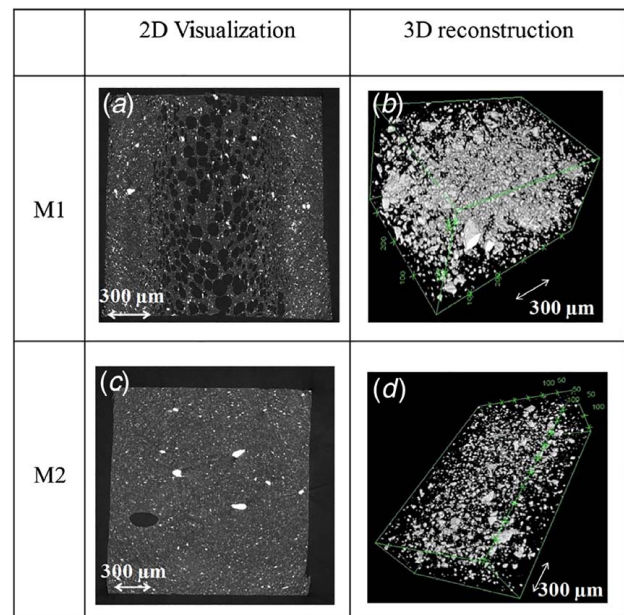


Fig. 11 X-ray μ -tomography images of PA composite with 10 wt% of MoS₂

matrix. From the tribological point of view, a pre-treatment with AMPTES and the improved dispersion with the matrix are not absolutely necessary because the presence of untreated solid lubricant filler in the sliding surface may deliver very good results. However, it could draw a conclusion that the second development methodology exerts a great good dispersion in the PA66 matrix and strong interfacial interaction between the modified surface of solid lubricant particles and PA66 matrix, resulting in greater enhancement of the micro-mechanical properties and the micro-structural aspects of the composites.

4 Conclusion

The present work was undertaken with an objective to explore the potential of the solid lubricant inclusions in PA-based composites and to study the friction and wear properties of the developed composites. This research work has focused on providing basic knowledge to encourage further research in this area. The analysis of the processing methodology and its dependence on the tribological behavior, scratch response, and filler dispersion leads to the following conclusions and recommendations:

- (1) Severe adhesive wear was observed on pure PA. The polymer exhibited a high friction coefficient and wear volume loss.
- (2) For composites injected molded directly, the incorporation of the two solid lubricant powders, particularly graphite, was beneficial in reducing the friction coefficient and the wear volume loss of PA.
- (3) Composite filled with 10 wt% graphite corresponded to the best tribological parameters. The adhesive and plowing wear was extremely reduced.
- (4) The second development methodology, including surface modification of the particles by AMPTES, mechanical mixing with the PA matrix, and microinjection, did not improve the friction and wear results.
- (5) The means performed to characterize the filler/matrix adhesion for the composites developed conventionally (M1) showed poor interfacial adhesion with debonding and fracture phenomena. Tomographic visualization in 3D showed filler agglomeration as well as the abundance of a large number of gas bubbles trapped in the composites.
- (6) The scratch test parameters were improved with the second development methodology particularly with the MoS₂ filler. SEM micrographs of the fracture exhibited an improved interfacial adhesion between the filler and the matrix. Debonding and voids were clearly reduced. X-ray tomography features showed homogeneous dispersion of the solid lubricant particles in the thermoplastic composites, a significant decrease in the density of the gas bubbles was also noticed.

Acknowledgment

The authors gratefully acknowledge the financial assistance of the Electro-Mechanical Systems Laboratory in the National Engineering School of Sfax funded by the Ministry of Higher Education and Scientific Research of Tunisia as well as the assistance provided by the Department of Applied Mechanics in the FEMTO-ST Institute at Besançon, France.

References

- [1] Erdemir, A., 2005, "Review of Engineered Tribological Interfaces for Improved Boundary Lubrication," *Tribol. Int.*, **38**(3), pp. 249–256.
- [2] Souza, J. C. M., Bentes, A. C., Reis, K., Gavinha, S., Buciumeanu, M., Henriques, B., Silva, F. S., and Gomes, J. R., 2016, "Abrasive and Sliding Wear of Resin Composites for Dental Restorations," *Tribol. Int.*, **102**(154), pp. 154–160.
- [3] Shalwan, A., and Yousif, B. F., 2013, "In State of Art: Mechanical and Tribological Behaviour of Polymeric Composites Based on Natural Fibres," *Mater. Design*, **48**, pp. 14–24.
- [4] Mazza, L., Trivella, A., Grassi, R., and Malucelli, G., 2015, "A Comparison of the Relative Friction and Wear Responses of PTFE and a PTFE-Based Composite When Tested Using Three Different Types of Sliding Wear Machines," *Tribol. Int.*, **90**, pp. 15–21.
- [5] Jiang, Z., Gyurova, L. A., Schlarb, A. K., Friedrich, K., and Zhang, Z., 2008, "Study on Friction and Wear Behavior of Polyphenylene Sulfide Composites Reinforced by Short Carbon Fibers and Sub-Micro TiO₂ Particles," *Compos. Sci. Technol.*, **68**(3–4), pp. 734–742.
- [6] Shiva Kumar, K., and Chennakesava Reddy, A., 2016, "Study on Reinforcement Materials for Nylon Matrix Composites—A Review," *J. Sci. Eng. Res.*, **7**(6), pp. 156–160.
- [7] Ben Difallah, B., Kharrat, M., Dammak, M., and Monteil, G., 2014, "Improvement in the Tribological Performance of Polycarbonate via the Incorporation of Molybdenum Disulfide Particles," *Tribol. T.*, **57**(5), pp. 806–813.
- [8] Ben Difallah, B., Kharrat, M., Dammak, M., and Monteil, G., 2012, "Microstructure, Friction and Wear Analysis of Thermoplastic Based Composites With Solid Lubricant," *Mech. Ind.*, **13**(5), pp. 337–346.
- [9] Ben Difallah, B., Kharrat, M., Dammak, M., and Monteil, G., 2012, "Mechanical and Tribological Response of ABS Polymer Matrix Filled With Graphite Powder," *Mater. Design*, **34**, pp. 782–787.
- [10] Lv, Y., Wang, W., Xie, G., and Lu, J., 2018, "Self-Lubricating PTFE-Based Composites With Black Phosphorus Nanosheets," *Tribol. Lett.*, **66**(2), p. 61.
- [11] Wang, W., Xie, G., and Luo, J., 2018, "Black Phosphorus as a New Lubricant," *Friction*, **6**(1), pp. 116–142.
- [12] Puértolas, J. A., Castro, M., Morris, J. A., Rios, R., and Ansón-Casaos, A., 2019, "Tribological and Mechanical Properties of Graphene Nanoplatelets/PEEK Composites," *Carbon*, **141**, pp. 107–122.
- [13] Zhang, L., Qi, H., Li, G., Zhang, G., Wang, T., and Wang, Q., 2017, "Impact of Reinforcing Fillers' Properties on Transfer Film Structure and Tribological Performance of POM-Based Materials," *Tribol. Int.*, **109**, pp. 58–68.
- [14] Bart, J. C. J., Gucciardi, E., and Cavallaro, S., 2013, *Biolubricants: Science and Technology*, 1st ed., J. C. J. Bart, E. Gucciardi, and S. Cavallaro, eds., Woodhead Publishing Ltd., Cambridge, pp. 24–27.
- [15] Rother, R., and DeArmitt, C., 2017, *Brydson's Plastics Materials*, 8th ed., M. Gilbert, ed., Elsevier, Butterworth-Heinemann, pp. 169–204.
- [16] Yeo, J.-S., Seong, D.-W., and Hwang, S.-H., 2015, "Chemical Surface Modification of Lignin Particle and Its Application as Filler in the Polypropylene Composites," *J. Ind. Eng. Chem.*, **31**, pp. 80–85.
- [17] Wu, X., Qiu, J., Liu, P., and Sakai, E., 2013, "Preparation and Characterization of Polyamide Composites With Modified Graphite Powders," *J. Polym. Res.*, **20**(11), pp. 284–291.
- [18] Jiang, H., Browning, R., Fincher, J., Gasbarro, A., Jones, S., and Sue, H.-J., 2008, "Influence of Surface Roughness and Contact Load on Friction Coefficient and Scratch Behavior of Thermoplastic Olefins," *Appl. Surf. Sci.*, **254**(15), pp. 4494–4499.
- [19] Jiang, H., Cheng, Q., Jiang, C., Zhang, J., and Lib, Y., 2015, "Effect of Stick-Slip on the Scratch Performance of Polypropylene," *Tribol. Int.*, **91**(1–5), pp. 1–5.
- [20] Bermúdez, M. D., Brostow, W., Carrión-Vilches, F. J., and Sanes, J., 2010, "Scratch Resistance of Polycarbonate Containing ZnO Nanoparticles: Effects of Sliding Direction," *J. Nanosci. Nanotechnol.*, **10**(10), pp. 6683–6689.
- [21] Wong, M., Moyse, A., Lee, F., and Sue, H.-J., 2004, "Study of Surface Damage of Polypropylene Under Progressive Loading," *J. Mater. Sci.*, **39**(10), pp. 3293–3308.
- [22] Jiang, H., Browning, R. L., Hossain, M. M., Sue, H.-J., and Fujiwar, M., 2010, "Quantitative Evaluation of Scratch Visibility Resistance of Polymers," *Appl. Surf. Sci.*, **256**(21), pp. 6324–6329.
- [23] Chu, J., Xiang, C., Su, H. J., and Damon Hollis, R., 2000, "Scratch Resistance of Mineral-Filled Polypropylene Materials," *Polym. Eng. Sci.*, **40**(4), pp. 944–955.
- [24] Moghbelli, E., Browning, R. L., Boo, W.-J., Hahn, S. F., Feick, L. J. E., and Su, H. J., 2008, "Effects of Molecular Weight and Thermal History on Scratch Behavior of Polypropylene Thin Sheets," *Tribol. Int.*, **41**(5), pp. 425–433.
- [25] Zokaie, S., Lesan Khoch, R. M., and Bagheri, R., 2007, "Study of Scratch Resistance in Homo- and Co-Polypropylene Filled With Nanometric Calcium Carbonate," *Mater. Sci. Eng. A*, **445–446**, pp. 526–536.
- [26] Feng, P., Zhang, C., Wu, Z., and Zhang, J., 2013, "Effect of Scratch Velocity on Deformation Features of C-Plane Sapphire During Nanoscratching," *J. Mech. Eng.*, **59**(6), pp. 367–374.
- [27] Wong, M., Lim, G. T., Moyse, A., Reddy, J. N., and Sue, H.-J., 2004, "A New Test Methodology for Evaluating Scratch Resistance of Polymers," *Wear*, **256**(11), pp. 1214–1227.
- [28] Jiang, H., Browning, R., and Sue, H.-J., 2009, "Understanding of Scratch-Induced Damage Mechanism," *Polymer*, **50**(16), pp. 4056–4065.
- [29] Purslow, D., 1981, "Some Fundamental Aspects of Composites Fractography," *Composites*, **12**(4), pp. 241–247.
- [30] Wang, H., Li, H., and Yan, F., 2005, "Reduction in Wear of Metakaolinite-Based Geopolymers Composite Through Filling of PTFE," *Wear*, **258**(10), pp. 1562–1566.
- [31] Kang, S. C., and Chung, D. W., 2003, "Improvement of Frictional Properties and Abrasive Wear Resistance of Nylon/Graphite Composite by Oil Impregnation," *Wear*, **25**(1–2), pp. 103–110.
- [32] Myshkin, N. K., Petrokovets, M. I., and Kovalev, A. V., 2005, "Tribology of Polymers: Adhesion, Friction, Wear, and Mass-Transfer," *Tribol. Int.*, **38**(11–12), pp. 910–921.
- [33] Zhang, X., Liao, G., Jin, Q., Feng, X., and Jian, X., 2008, "On Dry Sliding Friction and Wear Behavior of PPSK filled with PTFE and Graphite," *Tribol. Int.*, **41**(3), pp. 195–201.

- [34] Chen, Z., Li, T., Yang, Y., Liu, X., and Lv, R., 2004, "Mechanical and Tribological Properties of PA/PPS Blends," *Wear*, **257**(7), pp. 696–707.
- [35] Chang, L., Zhang, Z., Zhang, H., and Schlarb, A. K., 2006, "On the Sliding Wear of Nanoparticle Filled Polyamide 66 Composites," *Compos. Sci. Technol.*, **66**(16), pp. 3188–3198.
- [36] Byett, J. H., and Allen, C., 1992, "Dry Sliding Wear Behaviour of Polyamide 66 and Polycarbonate Composites," *Tribol. Int.*, **25**(4), pp. 237–246.
- [37] Samyn, P., De Baets, P., Schoukens, G., and Van Driessche, I., 2007, "Friction, Wear and Transfer of Pure and Internally Lubricated Cast Polyamides at Various Testing Scales," *Wear*, **262**(11–12), pp. 1433–1449.
- [38] Liu, L., Li, W., Tang, Y., Shen, B., and Hu, W., 2009, "Friction and Wear Properties of Short Carbon Fiber Reinforced Aluminum Matrix Composites," *Wear*, **266**(7–8), pp. 733–738.
- [39] Guo, Q. B., Rong, M. Z., Jia, G. L., Lau, K. T., and Zhang, M. Q., 2009, "Sliding Wear Performance of Nano-SiO₂/Short Carbon Fiber/Epoxy Hybrid Composites," *Wear*, **266**(7–8), pp. 658–665.
- [40] Xiang, C., Sue, H.-J., Chu, J., and Coleman, B., 2001, "Scratch Behavior and Material Property Relationship in Polymers," *J. Polym. Sci. B: Polym. Phys.*, **39**(1), pp. 47–59.
- [41] Phueakbuakhao, N., Prissanaroon-Ouajai, W., and Kreua-Ongarjnukool, N., 2008, "Effect of Coupling Agents on Mechanical Properties and Morphology of CaCO₃-Filled Recycled High Density Polyethylene," *J. Metals, Mater. Miner.*, **18**(2), pp. 131–135.
- [42] Deng, M., Tan, V. B. C., and Tay, T. E., 2004, "Atomistic Modeling: Interfacial Diffusion and Adhesion of Polycarbonate and Silanes," *Polymer*, **45**(18), pp. 6399–6407.